Causal Effect of Scholarships on Student Dropout & Academic Success

Machine Learning for Econometrics
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Research Question

Can we identify a **causal effect** of scholarship holding on students' academic success?

Dataset

Predict Students' Dropout and Academic Success
UCI ML Repository · 4,424 students
Polytechnic University of Portalegre, 2008-2018

Methods

Double Post-Lasso · DML-PLR · DML-IRM
AIPW · Heterogeneous Treatment Effects
Policy Learning · Placebo Tests

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Outline

1. Introduction & PICO
2. Exploratory Data Analysis
3. Causal Identification & DAG
4. Statistical Estimators & Model Selection
5. Robustness & Sensitivity Analysis
6. Conclusion & Discussion

1 Introduction & PICO

Motivation

Why scholarships?

- Literature shows a **positive effect** of need-based scholarships on degree acquisition, helping disadvantaged students close the attainment gap
- Scholarships are **not allocated at random**: they target low-income, high-performing students
  $\rightarrow$ many observable and unobservable **confounders**
- Identifying a *causal* effect requires methods that go beyond naive regression

Dataset at a Glance

Predict Students' Dropout and Academic Success

Source	UCI ML Repository
Students	4,424
University	Polytechnic of Portalegre
Period	2008-2018
Scholarship holders	$\approx$24%
Dropout rate	$\approx$32%

PICO Framework

Research Question

Can we identify a **causal effect** of scholarship holding on students' diploma attainment and/or dropout probability?

P - Population

4,424 students enrolled at the Polytechnic University of Portalegre between 2008 and 2018

C - Comparison

Students not awarded a scholarship, conditional on observable characteristics X

I - Intervention

Being awarded a scholarship at the time of enrollment in higher education

O - Outcome

- **Dropout** (left without degree)
- **Enrolled** (still within the degree)
- **Graduate** (obtained degree)

2 Exploratory Data Analysis

Variables Overview

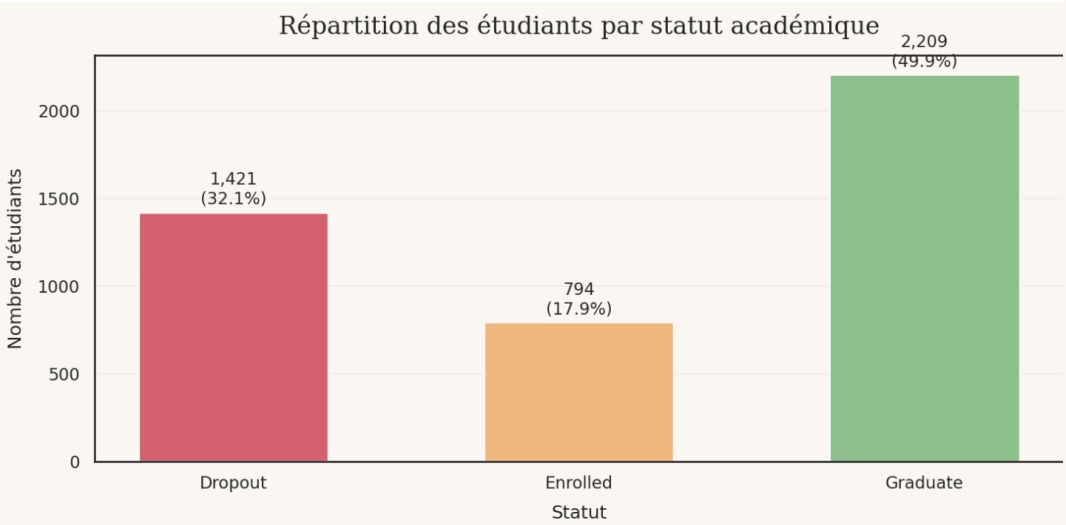
6 families of covariates

- **Demographic**: age, gender, marital status, nationality & displaced status
- **Socio-economic**: parents' education & occupation, special educational needs
- **Academic background**: programme, admission grade, application mode & rank
- **Financial**: debtor status, tuition fees up to date
- **Macroeconomic**: unemployment rate, inflation, GDP
- **Performance (S1 & S2)**: grades & credits
  → **post-treatment: excluded from the model**

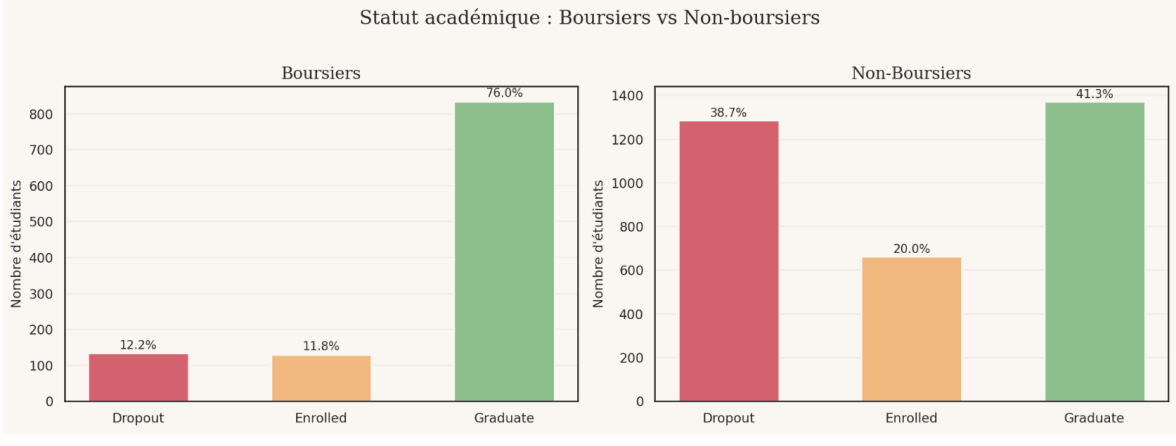
Key Figures

Observations	4,424
Scholarship holders	1,121
Dropout	32.1%
Graduate	50%
Enrolled	17.9%
Raw covariates	35
After dummy encoding	174
Missing values	0

Outcome Distribution



Outcome by Scholarship Status



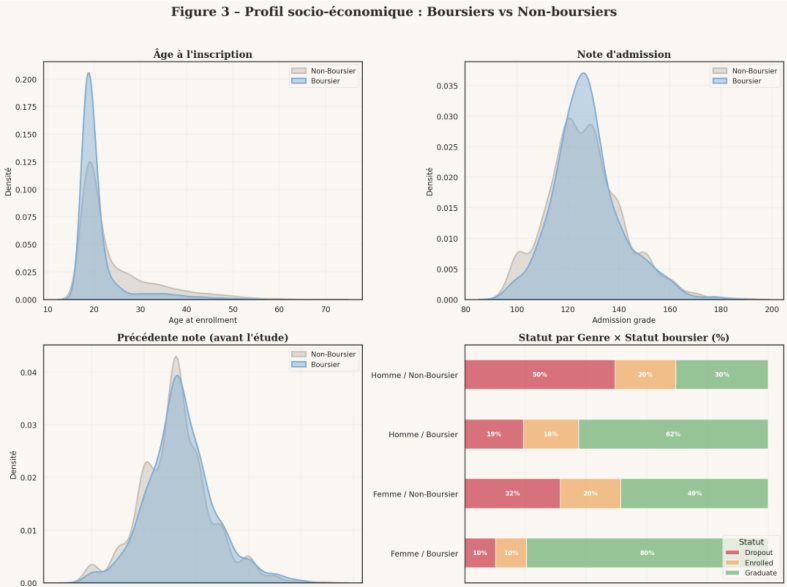
Outcome by Scholarship Status - Reading the Chart

- Scholarship holders have a **much lower dropout rate** and a **higher graduation rate** than non-holders
- The raw gap is striking : this is exactly what motivates the question

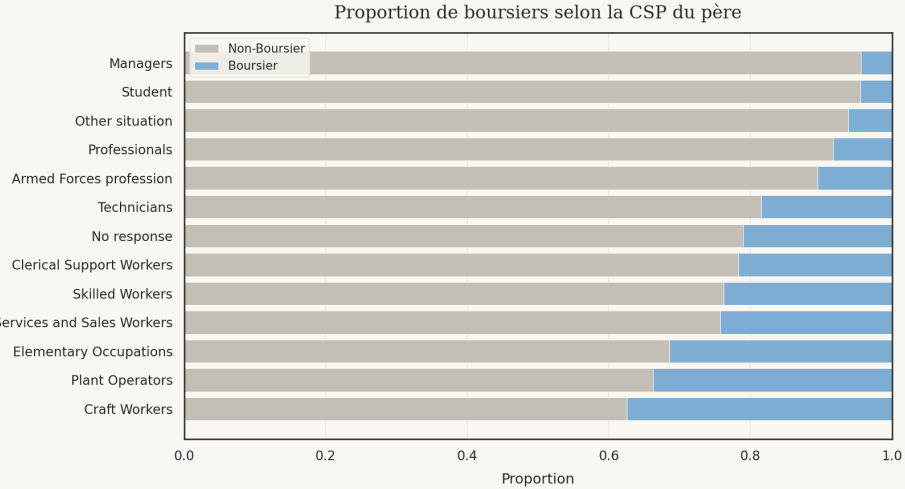
But this is NOT causal evidence

- Scholarship holders are a **selected group**: they tend to come from disadvantaged backgrounds but show strong academic potential - they may be more motivated
- The observed gap mixes the **causal effect** of the scholarship with **selection bias**

Socio-Economic Profile: Holders vs. Non-Holders

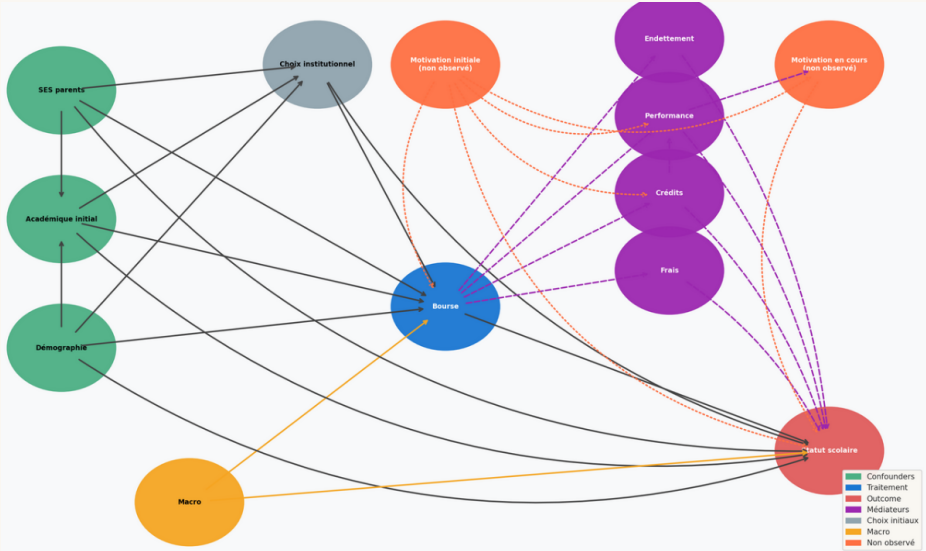


Holders by Father's Occupation



3 Causal Identification & DAG

Directed Acyclic Graph -



DAG - Reading the Graph

Node colour legend

- **Observed confounders**: SES, prior academics, demographics - these are the X we condition on
- **Treatment** D: scholarship holder
- **Outcome** Y and unobserved motivation U
- **Post-treatment** mediators: S1 grades, credits - caused by D, **must be excluded**
- **Macro** context

Key Assumption - CIA

Conditional Independence Assumption:

$$Y(0),\ Y(1) \perp D \mid X$$

Once we condition on all observed covariates X, the scholarship assignment is **as good as random**.

The DAG formalises *which* variables to include and which to exclude.

The unobserved node U (motivation) is the main threat. We cannot test CIA, but the DAG helps make the assumption explicit and discussable.

Identification Assumptions

1 - Unconfoundedness (CIA)

Conditional on X: $Y(0), Y(1) \perp D \mid X$

The DAG leaves no open back-door path between U and D once X is conditioned upon.

2 - Overlap (Positivity)

$\exists \epsilon > 0 : \epsilon < P(D{=}1 \mid X) < 1{-}\epsilon$

Every student must have a chance of being a holder *and* a non-holder. Verified via the propensity score histogram.

3 - SUTVA

No interference between students: one student's scholarship does not affect another student's outcome.

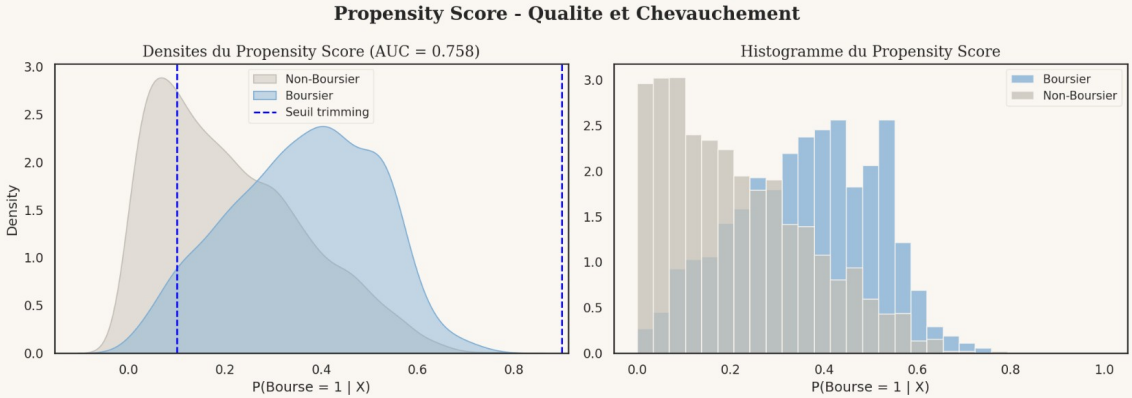
Reasonable individually, though peer effects could create mild violations.

Main Limitation

- **Unobserved**: intrinsic motivation, social capital
- **Single university**: limited external validity
- Post-treatment variables (S1 grades) **excluded** to avoid *bad control* bias

Assumption 1 (CIA) is *untestable* - it is a maintained hypothesis justified by the richness of our covariates and the DAG structure.

Propensity Score & Overlap Check



Propensity Score - Reading the Chart

What the chart shows

- The **left panel (KDE)** shows the estimated $\hat{e}(X) = P(D{=}1 \mid X)$ for scholarship holders (blue) and non-holders (grey)
- The **right panel (histogram)** shows the same distributions as counts
- The two curves **overlap substantially** - no region has zero support for either group

What to conclude

- The **overlap assumption is satisfied**: every student profile exists in both groups
- The moderate AUC confirms the propensity score is **predictive but not deterministic**
- Scores are trimmed to [0.01, 0.99] to stabilise IPW-based estimators (AIPW, IRM) at the tails

If the two distributions did not overlap, we could not compare the groups at all - the causal effect would not be identified. The overlap plot is therefore a *necessary diagnostic* before any causal analysis.

4 Statistical Estimators & Model Selection

Double Post-Lasso

Motivation

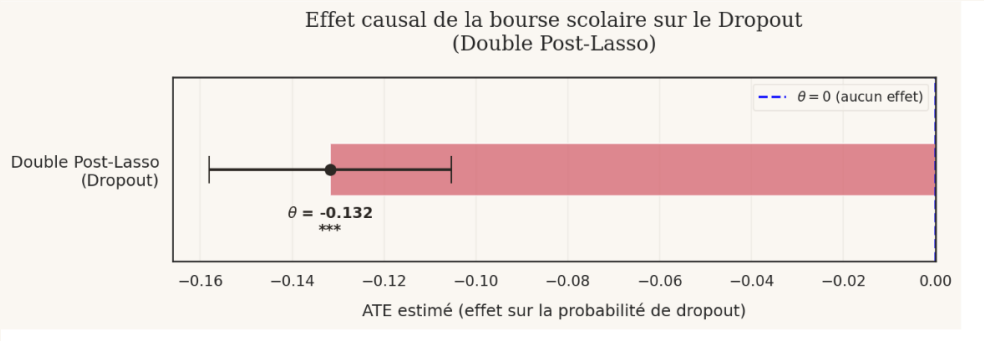
- After dummy encoding: We have a lot of features (with some sparse dummies)
- Inference with standard OLS could breaks down
- Lasso selects the relevant confounders; **Post-Lasso** OLS removes regularisation bias on the selected set

3-step procedure

1. Lasso Y on X $\rightarrow$ select features $\rightarrow$ logit Post-Lasso $\rightarrow$ residuals $\tilde{Y}$
2. Lasso D on X $\rightarrow$ select features $\rightarrow$ logit Post-Lasso $\rightarrow$ residuals $\tilde{D}$
3. OLS $\tilde{Y}$ on $\tilde{D}$ $\rightarrow$ $\hat{\theta}$

The key intuition is *partialling out* - we remove from both Y and D everything that can be explained by X, then regress what remains.

Double Post-Lasso - Result



Double Post-Lasso - Result

What the result tells us

- $\hat{\theta} \approx -0.10$ to -0.14: the scholarship reduces the probability of dropping out by roughly 10-14 percentage points
- The CI is entirely to the left of zero $\Rightarrow$ statistically significant
- This is our **baseline estimate** - we will check whether it holds under more flexible methods

This estimate is conditional on CIA. It is not from a randomised experiment - but it is our best estimate given the data.

Double Machine Learning - PLR

From Post-Lasso to DML

- Post-Lasso requires *sparsity*: what if the true $g(X)$ is non-linear or dense?
- DML replaces Lasso by **any ML algorithm** for nuisance estimation
- **Cross-fitting** ($K = 5$ folds, 2 reps) ensures the nuisance models are evaluated out-of-sample, removing regularisation and overfitting bias
- Same Neyman-orthogonal score $\Rightarrow$ $\sqrt{n}$-normality regardless of the learner

Three learners compared

- **Lasso**: linear sparse, fast
- **Random Forest**: non-parametric, handles interactions
- **Gradient Boosting**: low bias, flexible

Convergence across learners = **robustness test**

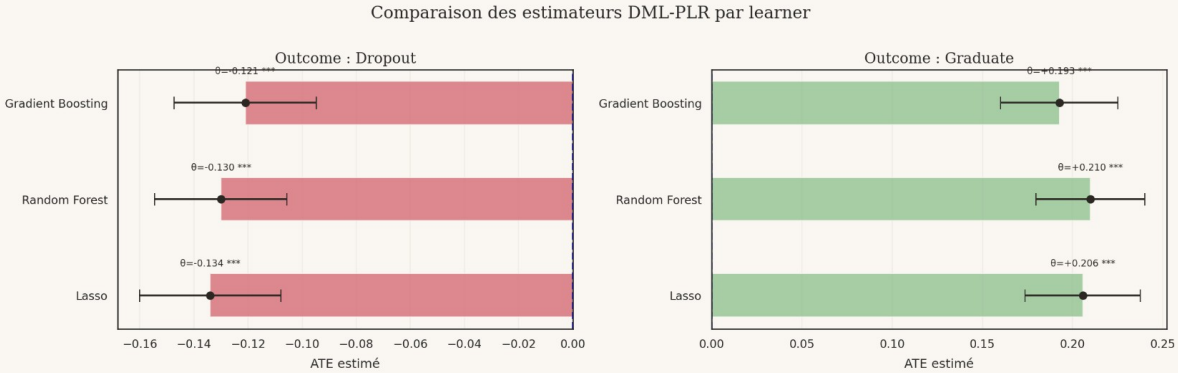
Why Cross-Fitting?

Without cross-fitting, the nuisance model would have *memorised* the training data. The residuals $\tilde{Y}$ and $\tilde{D}$ would be too small, biasing $\hat{\theta}$ toward zero.

Cross-fitting forces the model to generalise: nuisances are estimated on fold $-k$, residuals computed on fold k.

Cross-fitting and orthogonal score together are what makes the inference valid.

DML-PLR - Comparison Across Learners



DML-PLR Comparison - Reading the Chart

Key takeaway: robustness

- The estimates are **remarkably stable** across Lasso, RF and GB despite very different assumptions about the functional form of $g(X)$ and $m(X)$
- This convergence is our primary robustness argument: the result does not depend on the ML choice
- All CIs are entirely on the expected side of zero $\Rightarrow$ all significant at 5%

Which learner is best? it doesn't matter much here because they all converge - and that convergence is *itself* reassuring.

DML-IRM - Interactive Regression Model

Going beyond PLR

- PLR assumes **homogeneous treatment effect**: $Y = \theta D + g(X) + \varepsilon$
  Same θ for everyone
- IRM allows full $D \times X$ interaction:
  $Y = g_0(D, X) + \varepsilon$
  The effect of D can vary with X
- Target: **ATE** $= \mathbb{E}[g_0(1, X) - g_0(0, X)]$

Three nuisance functions

- $g_0(X) = \mathbb{E}[Y \mid D = 0, X]$
- $g_1(X) = \mathbb{E}[Y \mid D = 1, X]$
- $m_0(X) = P(D = 1 \mid X)$

`DoubleMLIRM` with Random Forest, trimming $= 0.01$

Orthogonal Score (AIPW form)

$$\psi_i = \hat{g}_1(X_i) - \hat{g}_0(X_i) + \frac{D_i(Y_i - \hat{g}_1(X_i))}{\hat{m}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{g}_0(X_i))}{1 - \hat{m}(X_i)}$$

Doubly robust: consistent if *either* $\hat{g}$ or $\hat{m}$ is correctly specified.

PLR vs. IRM

If the true model is IRM but we use PLR, we estimate a **weighted ATE** with overlap weights $\propto m(X)(1-m(X))$.

Equal to ATE only if effect is homogeneous or assignment is random.

AIPW Estimator & Heterogeneous Treatment Effects

AIPW - Doubly Robust

- Combines outcome model $\hat{\mu}_d(X)$ and propensity score $\hat{e}(X)$
- Consistent if *at least one* is correctly specified
- Achieves semiparametric efficiency bound
- Individual scores ψ_i reused for HTE analysis

Heterogeneous Effects (CATE)

- For each subgroup g: $\hat{\theta}_g = \frac{1}{|g|} \sum_{i \in g} \psi_i$
- *Generic ML* approach
- Subgroups: gender, age, debtor, fees status, attendance, mother's education

! The AIPW scores ψ_i are the "individual-level ATE estimates" - their average is the global ATE, and their subgroup averages are the CATEs.

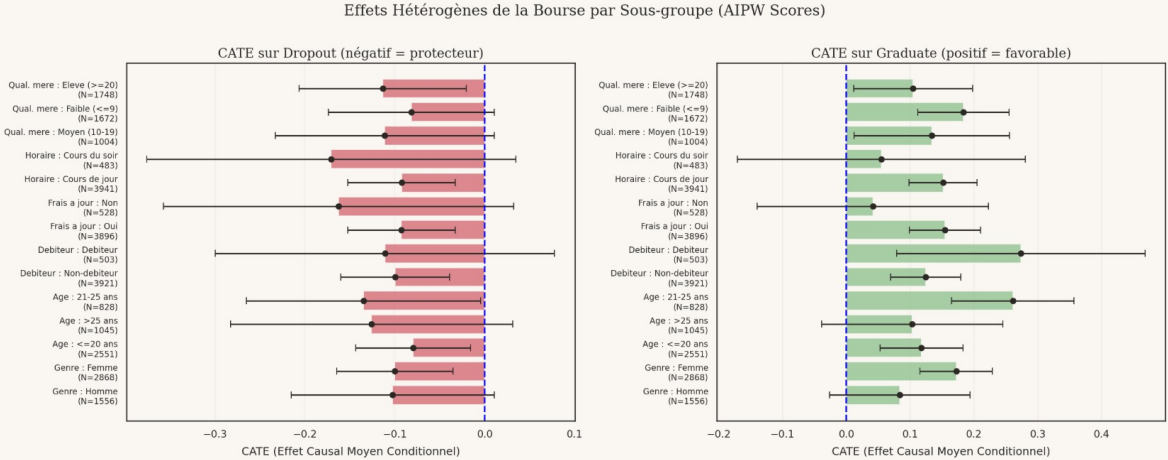
Why "Doubly Robust"?

The AIPW estimator has two components:

1. The **outcome model** part: $\hat{\mu}_1 - \hat{\mu}_0$
2. The **IPW correction**: adjusts for any residual bias in the outcome model

In practice: even if the RF slightly misspecifies g, $\hat{e}$ corrects for it.

Heterogeneous Treatment Effects by Subgroup



Heterogeneous Effects - Reading the Chart

Key findings

- **Debtors** and students **behind on fees** show the largest protective effect - financial stress amplifies the scholarship's impact
- Students **over 25** benefit less, possibly due to competing work commitments
- The effect is **consistent in sign** across all subgroups, but varies in magnitude

These are exploratory results - no correction for multiple testing. They suggest *where to look* for heterogeneity, not definitive conclusions.

Policy Learning - Optimal Scholarship Allocation

From estimation to policy

Rather than just measuring the average effect, we ask: **who benefits most from a scholarship?**

Welfare-maximising policy

$$V(\pi) = \mathbb{E}[\pi(X_i) \cdot \tau(X_i)]$$

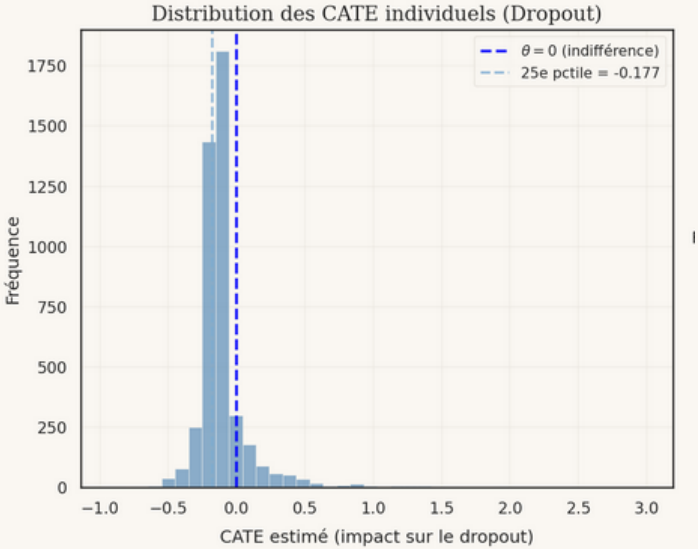
where $\tau(X_i)$ is the individual CATE.

Procedure

1. Train a **Random Forest** on AIPW scores ψ_i to predict $\hat{\tau}(X_i)$ for each student
2. Optimal policy: allocate scholarship to the **25% with the most negative** $\hat{\tau}$ (largest dropout-reduction effect)
3. Compare policy values: current policy vs. optimal vs. universal vs. none

The key idea is that the AIPW scores ψ_i are *unbiased proxies* for the individual causal effect. We can use them as a regression target to learn the optimal policy rule.

Policy Learning - CATE Distribution & Feature Importances



Policy Learning — Results & Interpretation

CATE Distribution & Importance

- **Heterogeneity:** The spread in predicted individual CATEs $\hat{\tau}(X_i)$ confirms that the scholarship effect varies significantly across students.
- **Drivers:** Financial stress (debtor status, tuition fees) and academic background are the primary features driving this heterogeneity.
- **Protection:** Most CATEs are negative, indicating the scholarship acts as a protective factor against dropout.

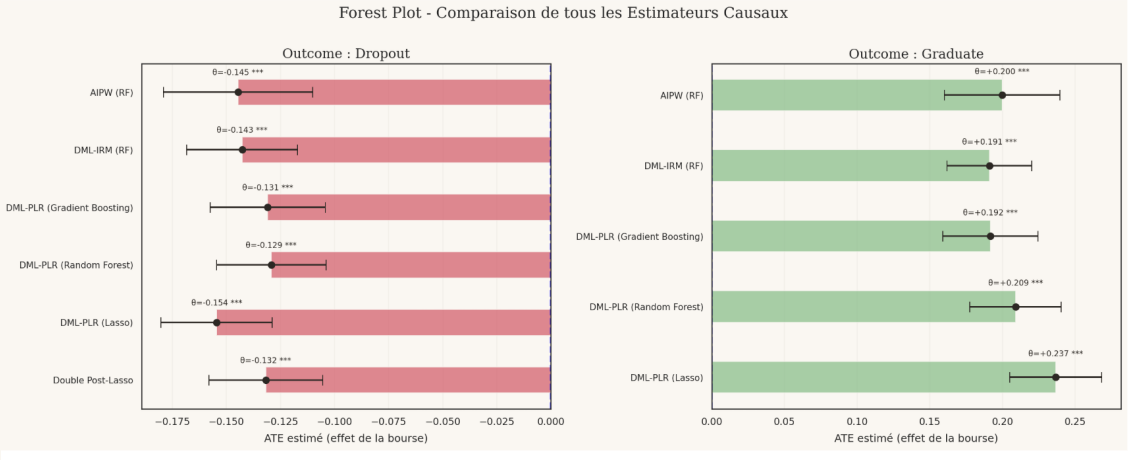
Expected Policy Values

Policy Type	Value (E[Y])
Current (Observed)	−0.0222
"Treat Everyone"	−0.1447
"Treat No One"	0.0000
sage!20 **Optimal (CATE-based)**	**−0.2054**

- *Note: Lower (more negative) is better (higher reduction in dropout).*

Key Takeaway: The optimal CATE-based policy outperforms the current status quo by **0.1832 points**, significantly reducing the expected dropout probability through better targeting.

All Causal Estimators - Final Forest Plot



All Estimators - Reading the Chart

The key message: convergence

- All six estimators produce **negative ATEs on Dropout** and **positive ATEs on Graduate**
- All CIs exclude zero at the 5% level
- The estimates are close in magnitude despite relying on very different assumptions

Methods compared

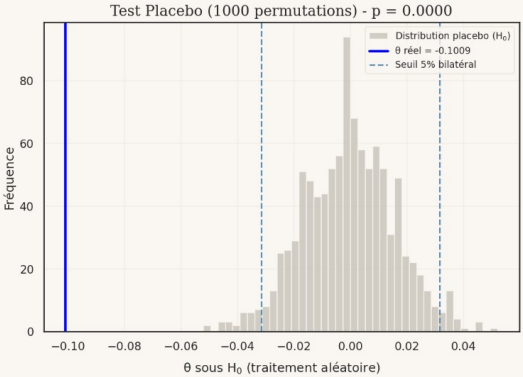
Method	Nuisance model	Allows HTE?
Post-Lasso / DML-PLR Lasso	Sparse linear	No
DML-PLR RF / GB	Non-parametric	No
DML-IRM RF / AIPW RF	Non-parametric, doubly robust	Yes

5 Robustness & Sensitivity Analysis

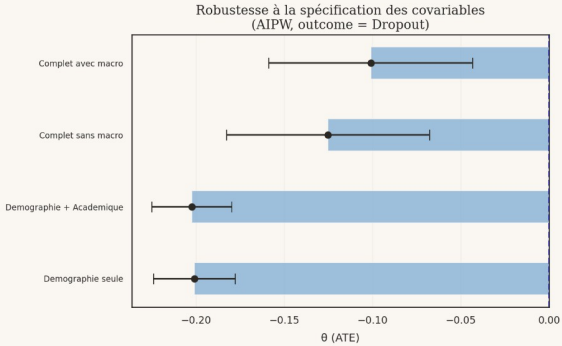
Placebo Test & Covariate Robustness

Placebo test

- Non-parametric validity check (Fisher, 1935).
- 1000 random permutations of scholarship assignment
- Under H_0 (no effect), the distribution is **centred at 0**
- The blue vertical line is our **real** $\hat{\theta} = \hat{\theta}_{(Id)}$
- **Far in the left tail** $\Rightarrow$ effect is not due to chance
- p-value < 0.05



Robustness Checks - Reading the Charts



Covariate robustness

- AIPW re-estimated with 4 different covariate sets
- **Stable estimates**: the result's sign does not depend on the choice of controls
- **Exception**: including S1 grades (post-treatment) biases $\hat{\theta}$ toward zero - this confirms they are mediators, not confounders

6 Conclusion & Discussion

Summary of Results

Main Finding

Receiving a scholarship **significantly reduces the dropout probability** and **increases the graduation rate**, with estimates robust across all methods.

Effect sizes (approximate)

- Dropout: ATE ≈ -0.10 to -0.14 (-10 to -14 pp)
- Graduate: ATE $\approx +0.10$ to $+0.14$ ($+10$ to $+14$ pp)
- All estimates significant at the 1% level

Heterogeneity

Debtors and students **behind on fees** benefit most. Students **over 25** benefit less (work-study conflict).

Policy Implication

Redirecting scholarships toward students with the highest estimated CATE would improve the **efficiency** of the allocation policy without increasing the total budget.

Main Limitations

- CIA is **untestable** (unobserved motivation)
- Single Portuguese university: limited external validity
- Binary treatment: no dose-response analysis
- Potential SUTVA violation via peer effects

Discussion & Perspectives

Key methodological lessons

- **DML separates** nuisance prediction (ML step) from causal estimation (orthogonal score step)
- **Cross-fitting** avoids regularisation and overfitting bias in nuisance estimation
- **Convergence across methods** is the best available robustness check in observational data
- **Post-treatment variables** must be excluded to avoid closing causal pathways (*bad controls*)

Possible extensions

- **Regression discontinuity**: exploit eligibility cutoffs for scholarship awards
- **Panel data + DiD**: if repeated cohort data become available
- Extend to multiple universities to improve external validity

Five Questions for the Audience

1. **[PICO / Identification]**
   Why can't we simply compare the dropout rates of scholarship holders and non-holders to estimate the causal effect? What does the DAG tell us about the confounders at play?

2. **[Double Post-Lasso]**
   What is the purpose of running a Lasso regression separately on Y and D before computing the final OLS estimate?

3. **[DML & Cross-Fitting]**
   Why is **cross-fitting** essential in Double Machine Learning? What bias would arise if the nuisance models and the causal estimate were computed on the same sample?

4. **[External validity]**
   What would be the next steps to improve the external validity of this study ?

5. **[Robustness]**
   What does the placebo test demonstrate ?

Thank you for your attention

Questions & comments welcome